

Qatar-10b

R-0149

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250824 · created 2026-07-16T09:15:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460911.66855 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.013
Transit depth (Rp/Rs)^2	3.01 +/- 0.45 [%]
Orbital Inclination (inc)	82.58 +/- 0.53
Airmass coefficient 1 (a1)	0.491 +/- 0.013
Airmass coefficient 2 (a2)	0.59 +/- 0.2
Scatter in the residuals of the lightcurve fit is	2.62 %
Best Comparison Star	#1 - [505, 343]
Optimal Method	PSF photometry
Transit Duration (day)	0.06464891950483319 +/- 1.4e-17

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.173 ± 0.013 vs 0.1265 (deviation 0.6σ)
Duration fit vs published	0.06464891950483319 d vs 0.1154 d (deviation 608230066155095.1σ)
Mid-time O-C	-32.3 min
Depth SNR	2.2
Residual scatter	2.62 %

Light curve

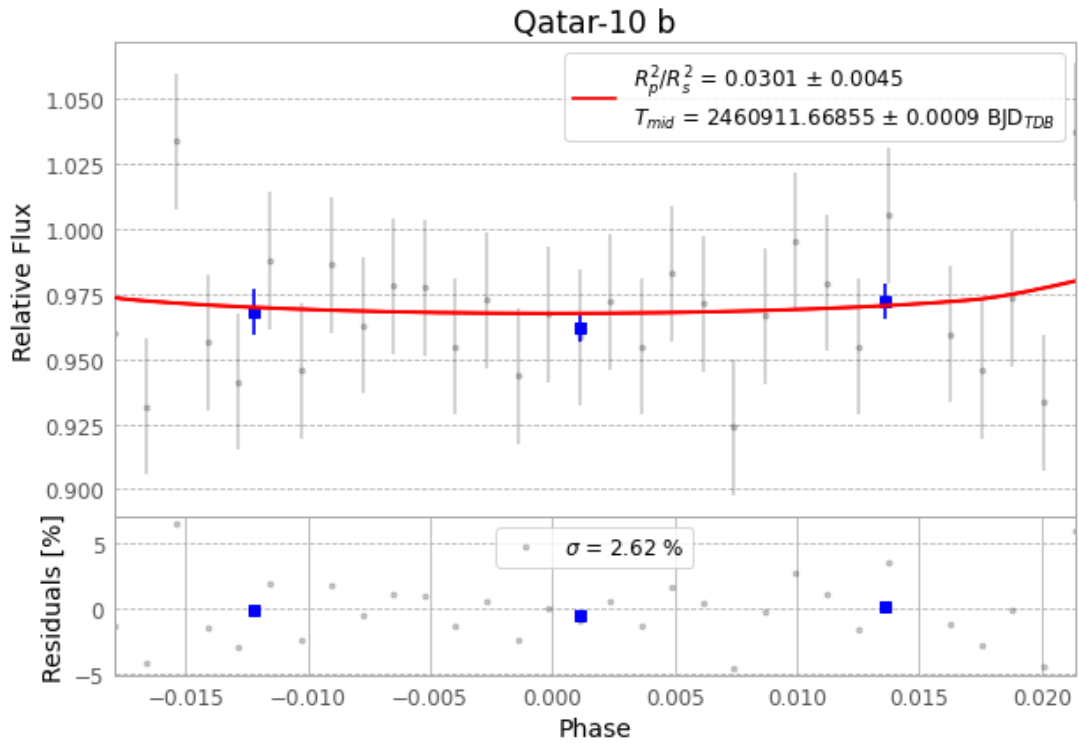


Figure 1 — FinalLightCurve_Qatar-10 b_2025-08-23.png (SHA-256 622a50d3c3c0bdda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 eed28b773fe9723f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

32 FITS frames (21 MB), Qatar-10250824031815.FITS → Qatar-10250824045115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250824.log (5d71f7a81840...), FinalLightCurve_Qatar-10 b_2025-08-23.png (622a50d3c3c0...), FinalLightCurve_Qatar-10 b_2025-08-23.csv (e87eb48812dd...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 09:15Z.